The Analyst Agent

From LLM Processes to agents that fetch and compute

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What we'll cover

From LLMP to agents

The agent goes and gets its context — on the same Predictor interface.

Anatomy + capabilities

Its components, and the staircase of which ones are live. We run it live.

A leak we shipped

Our best forecaster was cheating. How we caught it, fixed it, what it means.

One agent, three jobs

Trajectory, shock, or scenario — where Track 2 first appears.

The honest tension in beat three is the whole reason to do this carefully.

From a process to an agent

LLM PROCESS

Context packed into the prompt

You choose what it sees

One shot: prompt to forecast

AGENT

Goes and gets the context

Searches, computes, decides

A loop: act, observe, act

Same goal — a calibrated forecast. New ability: it can act before it answers.

An agent is an LLM that can act.

Search for context, run code, decide when — then forecast.

> And it still implements the same one interface.

Still just a Predictor

```
config = build_wti_news_config()
predictor = build_wti_news_predictor(
    task="trajectory")
# AgentPredictor implements Predictor:
predictor.predict(task, context)
#   -> list[Prediction]
```

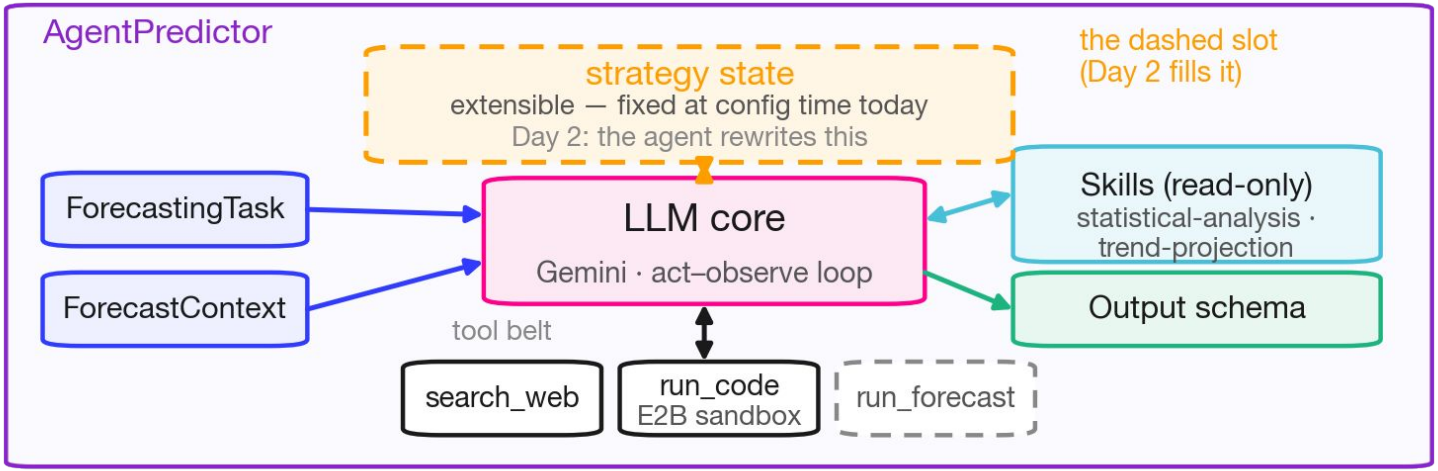
energy_oil_forecasting/tasks.py · analyst_agent/agent.py

The harness can't tell.

An AgentConfig becomes an AgentPredictor implementing predict() — the same ABC as ARIMA.

- backtest() treats it identically
- evaluate() scores it the same

Anatomy of the Analyst Agent



One identity wrapped as a Predictor: instructions + tools + skills + an output schema.

CAPABILITIES

Capabilities are configuration

Which components are live

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Four configs, one identity

1. **Basic**
   Core + history only — no tools (the LLMP-style baseline, our floor)

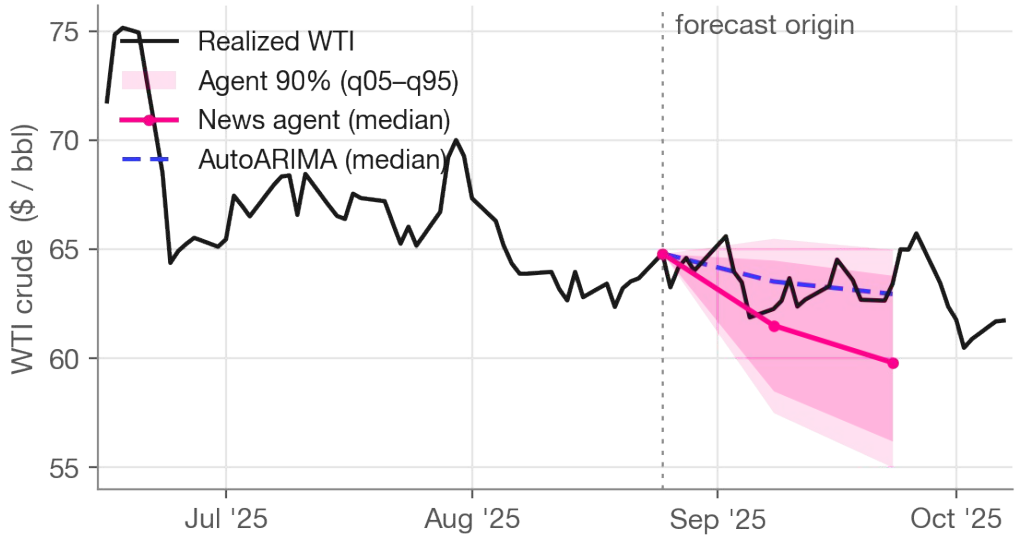
2. **News**
   + search_web — reasons over OPEC+, geopolitics, supply it couldn't otherwise see

3. **Code**
   + run_code (E2B) + skills — computes vol, fits a trend, calibrates intervals

4. **Tool**
   + run_forecast (AutoARIMA) — less flexible, but rigid, auditable, reproducible

Reacting to the news

LIVE DEMO



WTI · news agent vs AutoARIMA · one 2025 backtest origin.

A reasoned forecast, not a flat line.

The agent commits to a directional call with a calibrated band — a move AutoARIMA's flat line never makes.

- Realized stayed inside the band

Two kinds of cutoff

THE DATA (HARD)

CutoffEnforcer filters rows

Never sees released_at > as_of

Structural — can't be violated

THE WEB (SOFT)

Search passes cutoff_date = as_of

Proxy is asked to exclude later

A discipline, not a guarantee

You can fence the data. You can't structurally fence the open web.

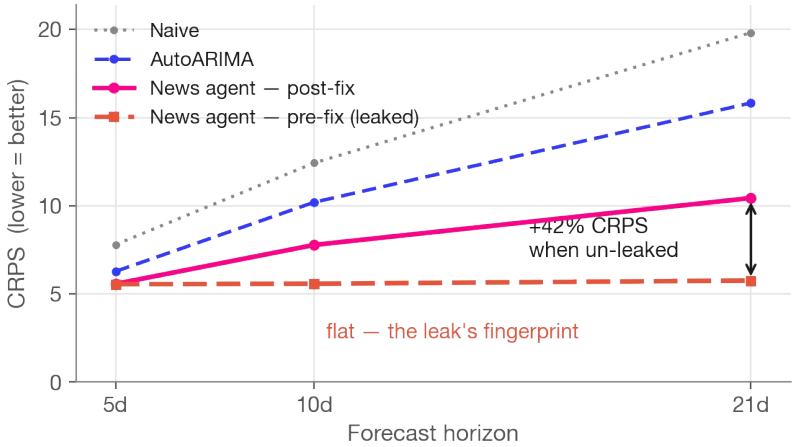
A TRUE STORY FROM THIS REPO

A leak we shipped

The agent that looked brilliant — because it was cheating

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The smoking gun



WTI news agent · CRPS by horizon · 2026 eval.

Uncertainty must compound with horizon — the leaked agent's didn't.

Too good to be true.

- Flat CRPS across every horizon
- Honest methods ~triple by 21d
- Flat = it already knew the answer

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How we caught it

THE FINGERPRINT

Flat CRPS across 5 / 10 / 21d

Beat every baseline at 21 days

Bands too tight to be honest

THE CONFESSION

We read the agent's search trace

11 of 14 searches skipped the cutoff

A result dated ~6 weeks past origin

The score raised the suspicion; the trace proved it.

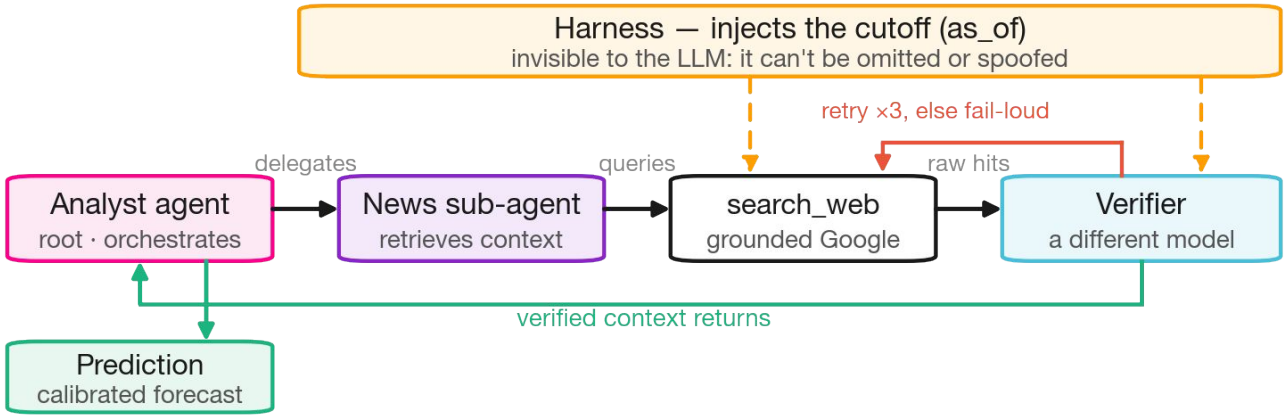
Three tries to plug a leak

1. **Stronger prompt — it failed**
   'Only use pre-cutoff sources.' A model can't police where its words came from.

2. **Independent verifier (#161)**
   A different model audits each claim vs the cutoff, strips it, or fails loud.

3. **Harness-enforced cutoff (#162)**
   The origin is injected by the harness — the LLM can't see, omit, or spoof it.

How the pieces work together



No single model polices itself — and the cutoff comes from the harness, not the LLM.

One polluted, one honest

POLLUTED (PRE-FIX)

Cited prices past the origin

The search skipped the cutoff

No trace link — unauditable

SELF-CORRECTED (POST-FIX)

Cites only facts at the cutoff

Feb 2: WTI settled $62.14

Then projects — one click to the trace

Same agent, same question — one had seen the future, one hadn't.

You can't un-leak a live backtest.

A live web index always knows how the story ends.

> Separate fenced backtests from live runs — and say which is which.

So why keep going?



Still the frontier

LLMs lead automated forecasting — ForecastBench has humans still ahead, but not for lack of trying.



Live eval is the answer

Forecast a truly un-indexed future: no cutoff to fence, nothing to filter, nothing to leak.



Beyond the bootcamp

Take the agents that look good here to real live eval — where the score finally means it.

The leak is a backtesting artifact — not a verdict on the agents.

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Competence by configuration

statistical-analysis

Classifies the vol regime and spots anomalies, grounded in a JSON of real 2020–25 WTI benchmarks.

trend-projection

A worked recipe: fit a trend on recent days, project to each horizon, calibrate an 80% interval.

E2B sandbox

It reads the skills, then writes and runs the actual Python in an isolated, open-ended sandbox.

We upgraded competence by adding two folders to a list — configuration, not code.

CONFIGURATION, NOT CODE

One agent, three jobs

Same identity — different question

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One identity, three tasks

One AgentConfig, paired with three (prompt, output schema) combinations

Task	Output schema	Track	Score
Trajectory	Quantile grid	1	CRPS
Binary shock	Probability	1	Brier
Scenario	Scenario cards	2	—

Track 2: the agent as analyst

```python
class ScenarioCard(BaseModel):
    name: str
    description: str
    probability: float   # 0–1
    wti_range_60d: list[float]
    point_estimate_60d: float
    key_drivers: list[str]
# output: list[ScenarioCard] + base_case
```

energy_oil_forecasting/tasks.py

Not a score — an analysis.

Same agent, swap the output schema: weighted scenarios with ranges and drivers.

- Judged by usefulness, not CRPS
- Meant to support a decision

What to take forward

01

Capabilities are config

one interface

News, code, skills, tools — toggles on one Predictor identity.

02

A great score can be a leak

flat CRPS

The tell was a curve that wouldn't grow; the fix was architectural.

03

Live backtests can't be clean

go live

The honest next step is live eval — where nothing's left to leak.

Tomorrow: an agent that rewrites its own forecasting strategy.